

Software Re-Engineering (SE4001)

Sessional-II Exam

Date: April 9, 2025

Course Instructor(s)

Dr. Farooq Ahmed

Total Time: 1 hour

Total Marks: 30

Total Questions: 3

Roll No

Section

Student Signature

Do not write below this line

Attempt all questions on the answer sheet

CLO 3: Apply different approaches to understand code

Question 1

[10+5 marks]

(a) SQL is the standard language used to query relational databases. An update query is used to modify a list of records and generally uses a table name, list of columns and their values and an optional condition (where clause). A few examples are given below:

UPDATE Customers SET ContactName = 'Ali Hamza', City = 'Lahore' WHERE CustomerID = 10	UPDATE Employee SET Status = 'Permanent' WHERE EmployeeCode = 5000	UPDATE Product SET Discount = 0
--	--	--

Write BNF for a general SQL update query as illustrated in the examples above. Note that WHERE clause is optional and it is assumed for simplicity that when used in a query it comprises of a single condition only.

(b) Javadocs is a common tool to generate source code documentation in readable format. Explain the process involved in understanding and conversion of code that Javadocs may employ.

CLO 5: Implement refactoring strategies for effective re-engineering

Question 2

[15 marks]

Consider a program to maintain current score for a tennis game. Scores for a particular game (not a set) can be calculated using following rules:

- A game is won by the first player to have won at least four points in total and at least two points more than the opponent.
- The running score of each game is described in a manner peculiar to tennis: scores from zero to three points are described as **Love**, **Fifteen**, **Thirty**, and **Forty** respectively.
- If at least three points have been scored by each player, and the scores are equal, the score is **Deuce**.

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- If at least three points have been scored by each side and a player has one more point than his opponent, the score of the game is **Advantage** for the player in the lead.

```
public class TennisGame {

    private int p2;
    private int p1;
    private String p1N;
    private String p2N;

    public TennisGame(String p1N, String p2N) {
        this.p1N = p1N;
        this.p2N = p2N;
    }

    public String getScore() {
        String s;
        if (p1 < 4 && p2 < 4 && !(p1 + p2 == 6)) {
            String[] p = new String[]{"Love", "Fifteen", "Thirty", "Forty"};
            s = p[p1];
            return (p1 == p2) ? s + "-All" : s + "-" + p[p2];
        } else {
            if (p1 == p2)
                return "Deuce";
            s = p1 > p2 ? p1N : p2N;
            return ((p1-p2)*(p1-p2) == 1) ? "Advantage " + s : "Win for " + s;
        }
    }

    public void wonPoint(String playerName) {
        if (playerName == "player1")
            this.p1 += 1;
        else
            this.p2 += 1;
    }

}
```

- (a) Analyze the code and list the **most relevant refactorings** applicable
- (b) Show **refactored code**.